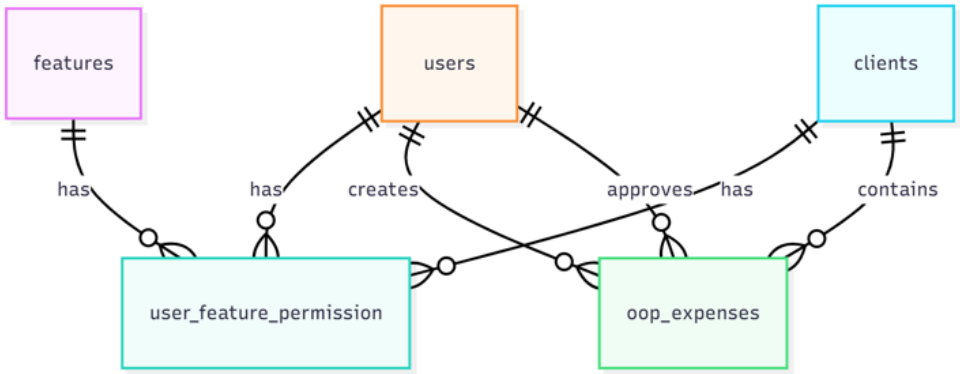
OOP User Management System

```
Primary Admin (X)
    │
    ├── Full access to all users (A, B, C, Y, Z...)
    │
    └── Can delegate management rights to → Admin User (Y)
            │
            └── Manages subset of users (A, B, C...)
```

```
erDiagram
    features ||--o{ user_feature_permission : has
    users ||--o{ user_feature_permission : has
    clients ||--o{ user_feature_permission : has
    users ||--o{ oop_expenses : creates
    clients ||--o{ oop_expenses : contains
    users ||--o{ oop_expenses : approves
```



example permission metrics:

```
{
  "role": "Primary Administrator",
  "default_permissions": {
    "own_expenses": ["create", "view", "update", "delete", "approve"],
    "all_users_expenses": ["create", "view", "update", "approve"],
    "management_rights": ["grant", "revoke"]
  }
}
{
  "role": "Administrator",
  "default_permissions": {
    "own_expenses": ["create", "view", "update", "delete", "approve"],
    "all_users_expenses": [],
    "management_rights": []
  },
  "with_management_rights": {
    "managed_users_expenses": ["create", "view", "update", "approve"],
```

```
    "management_rights": ["grant", "revoke"]
  }
}
{
  "role": "Business User",
  "default_permissions": {
    "own_expenses": ["create", "view", "update", "delete"],
    "all_users_expenses": [],
    "management_rights": [],
    "approval_rights": []
  },
  "with_management_rights": {
    "managed_users_expenses": ["create", "view", "update"],
    "approval_rights": []  // Cannot approve even managed users' expenses
  }
}
{
  "role": "Card User",
  "default_permissions": {
    "own_expenses": ["create", "view", "update", "delete"],
    "all_users_expenses": [],
    "management_rights": [],
    "approval_rights": []
  },
  "with_management_rights": {
    "managed_users_expenses": ["create", "view", "update"],
    "approval_rights": []  // Cannot approve even managed users' expenses
  }
}
```

```sql
CREATE TABLE user_feature_permission (
    id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    user_id BIGINT UNSIGNED NOT NULL,
    client_id BIGINT UNSIGNED NOT NULL,
    feature_id BIGINT UNSIGNED NOT NULL,
    grantor_id BIGINT UNSIGNED NOT NULL,
    manager_user_id BIGINT UNSIGNED NULL,
    is_enabled BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP NULL,
    updated_at TIMESTAMP NULL,
    UNIQUE KEY unique_user_feature (user_id, client_id, feature_id),
    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE,
    FOREIGN KEY (client_id) REFERENCES clients(id) ON DELETE CASCADE,
    FOREIGN KEY (feature_id) REFERENCES features(id) ON DELETE CASCADE,
    FOREIGN KEY (grantor_id) REFERENCES users(id) ON DELETE CASCADE,
    FOREIGN KEY (manager_user_id) REFERENCES users(id) ON DELETE CASCADE
);
```

Example data

```sql
-- Client A (ID: 1) - Complete user_feature_permission entries
INSERT INTO user_feature_permission
(user_id, client_id, feature_id, grantor_id, manager_user_id, is_enabled, created_at) VALUES

-- Primary Admins with management capabilities
(101, 1, 16, 100, NULL, TRUE, '2024-01-15 09:00:00'),

-- Admins with management capabilities
(102, 1, 16, 101, NULL, TRUE, '2024-01-15 09:00:00'),  -- Admin 102 enabled
(103, 1, 16, 101, NULL, TRUE, '2024-01-15 09:00:00'),  -- Admin 103 enabled
(104, 1, 16, 101, NULL, TRUE, '2024-01-15 09:00:00'),  -- Admin 104 enabled (no management)

-- Card Users with managers assigned
(105, 1, 16, 102, 109, TRUE, '2024-01-15 09:00:00'),
(105, 1, 16, 101, 102, TRUE, '2024-01-15 09:00:00'),-- Managed by Admin 102
```

```
(106, 1, 16, 101, 102, TRUE, '2024-01-15 09:00:00'),  -- Managed by Admin 102
(107, 1, 16, 101, 102, FALSE, '2024-01-15 09:00:00'),  -- Managed by Admin 102 - revoked
(108, 1, 16, 101, 103, TRUE, '2024-01-15 09:00:00'),  -- Managed by Admin 103

```

Keep an audit log of granting, enable and disable of permissions

```
CREATE TABLE oop_expenses (
    id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    user_id BIGINT UNSIGNED NOT NULL,
    client_id BIGINT UNSIGNED NOT NULL,
    date DATE NOT NULL,
    merchant_name VARCHAR(255) NOT NULL,
    description TEXT NULL,
    transaction_type ENUM('atm_withdrawal', 'point_of_sale', 'fee_charges', 'refund') NOT NULL,
    currency VARCHAR(3) NOT NULL,
    amount DECIMAL(10,2) NOT NULL,
    merchant_address TEXT NULL,
    country VARCHAR(100) NULL,
    source VARCHAR(100) NOT NULL,
    category VARCHAR(100) NULL,
    custom_fields JSON NULL,
    tracking_code_i VARCHAR(100) NULL,
    tracking_code_ii VARCHAR(100) NULL,
    project_id VARCHAR(100) NULL,
    vat DECIMAL(10,2) NULL,
    status ENUM('pending', 'approved', 'rejected', 'in_review') DEFAULT 'pending',
    approved_by BIGINT UNSIGNED NULL,
    approved_at TIMESTAMP NULL,
    receipt_path VARCHAR(500) NULL,
    notes TEXT NULL,
    created_at TIMESTAMP NULL,
    updated_at TIMESTAMP NULL,
    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE,
    FOREIGN KEY (client_id) REFERENCES clients(id) ON DELETE CASCADE,
    FOREIGN KEY (approved_by) REFERENCES users(id) ON DELETE SET NULL
);
```

5. Access Control Flow

Service Enablement Flow:

1. **Sales/Ops Management** enables Expense via Admin.Volopa
2. System validates client has card services active
3. Email notification sent to client Primary Admin
4. Default permission applied automatically

1. Default Permission Once service is enabled

ALL USERS (card user, business user, administrator, primary administrator) under the client can

- Add own expense
- Edit own expense
- View own expense

Administrator can do all above and

- Approve own expense

Primary Administrator can do all above and

- Add expense for All
- Edit expense for All
- View expense for all
- Approve expense for all
- Grant access to others

2. Granting Managing rights

Primary Administrator and Administrator can grant access via: Web App: Settings > Manage Users.

Volopa Admin(1) can grant access via Admin.Volopa: User accounts page

Managing access can be given to any user irrespective of the role

3. Managing user Once Managing rights are granted

Any user under a client can be given managing rights to another user

Card user and Business user once get managing right of another user, they can

- Add expense for the managed user
- Edit expense for the managed user
- View expense for the managed user

Administrator can do all above and

- approve expense
- Can grant access to others(can only grant access to managed users)

4. Revoking managing right

Primary Administrator can revoke access via: Web App: Settings > Manage Users.

Volopa Admin(1) can revoke access via Admin.Volopa: User accounts page

If managing right is revoked, the user loosed access to add, edit, view / approve access to that specific user. Until another manager is set the base user is managed by Primary administrator.